

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

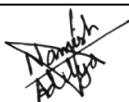
Project Abstract Submission Form

Sec No- 10	Team No-	Project Title- Campus Connect: Event Aggregator & Manager
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1. Abstract

Campus Connect is an event aggregation and management platform designed to address the difficulty students face in discovering timely, relevant university activities and the difficulty administrators face in maintaining reliable event information. The proposed system provides a public event catalogue with search, category filtering, pagination, event details, and upcoming-event recommendations. Administrators receive a protected control panel for creating, updating, deleting, filtering, and analysing events, with CSV export for reporting. Students can record interest in an event through a transaction-safe workflow while the configured external registration link remains authoritative for actual registration. The platform also supports validated event-image uploads, role-based access control, secure sessions, rate limiting, audit logging, health monitoring, and metrics. Its modular monolith architecture uses a relational MySQL database with JPA and versioned Flyway migrations, providing strong data integrity, query performance, and a practical foundation for future distributed service extraction.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090104	Tejaswin Amara	
2520090110	Namish Aditya Vallala	

3. Frameworks / Platforms / Software Used

- Java 25 and Maven Wrapper
- Spring Boot 4.1.0 with Spring MVC
- Thymeleaf for server-rendered web pages
- Spring Security for authentication, authorization, CSRF protection, and secure sessions
- Spring Data JPA / Hibernate for persistence
- MySQL 8.4 LTS as the relational database
- Flyway for versioned database migrations
- Spring Boot Actuator and Micrometer Prometheus Registry for health and metrics
- Resilience4j and Bucket4j for resilience and rate limiting
- Docker and Docker Compose for containerized deployment
- GitHub Actions for continuous integration, testing, and quality checks

Guide Name and Signature (remarks if any)